Software Product

C.A.R.E. = Car Alert & Reminder Eye

"An extra eye watching the **back seat** for you."

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introduction - C.A.R.E

Car Alert & Reminder Eye

Project Goal: To develop a smart, standalone automotive safety system designed to prevent fatalities and injuries resulting from children or pets being forgotten in closed vehicles. The system provides an "extra eye" for parents, operating passively without requiring active activation or any change in driving habits.

Specific Objectives:

1. **Reliable and Accurate Detection:** Creating a system that combines image processing and breathing sensors to positively identify the presence of a living being in the vehicle.
2. **Full Autonomy:** Developing logic that automatically detects the end of the drive (vehicle engine off) and activates scanning without user intervention.
3. **Real-Time Alerting:** Building a multi-channel alert system: a vocal alert in the vehicle's surroundings and an immediate notification to the driver's mobile device (SMS/Push).
4. **Simplicity and Accessibility (Plug & Play):** Designing a product that does not require professional installation or connection to the vehicle's computer, but connects to a standard power source (USB) and operates independently.

Problem Presentation: Forgotten Baby Syndrome (FBS) is a tragic occurrence, often occurring due to a change in routine, Exhaustion, or momentary distraction of the parent, and not as a result of intentional neglect. The temperature in a closed vehicle rises rapidly to lethal levels, and the danger of death or brain damage exists after just a few minutes.

Despite the severity of the problem, existing solutions in the market today are lacking. Built-in systems exist only in new and expensive models, and often provide a generic "Check Back Seat" alert even when the vehicle is empty, causing drivers to ignore them. Other solutions often require complex installation, connection to the vehicle's computer (OBD), or reliance on an app that requires the parent to actively activate it at the start of every trip. There is a vital need for a protection system that does not require the user to remember to activate it and can be simply added to any existing vehicle.

Solution: C.A.R.E - Car Alert & Reminder Eye A product that serves as an "extra eye" protecting the vehicle's Passengers. The system functions as a standalone unit connecting to the existing USB connection in the vehicle, with no need for professional installation.

How the System Works: The system detects the power cut in the lighter socket. At this moment, the system switches to operate on an internal backup battery and enters "scan" mode. The system activates an algorithm combining two input sources: visual detection and motion detection. If a living being is detected in the vehicle, a vocal alarm will be triggered in the vehicle's surroundings, and an urgent message will be sent to the driver's mobile device with a clear indication of the danger. If nothing is detected after a short scan, the system shuts down completely.

This solution meets the workshop requirements by providing a full technological response to a painful human problem, using advanced technologies.

System Characterization - C.A.R.E

(Car Alert & Reminder Eye)

1. General Description and System Goals

The C.A.R.E. system is a life-saving IoT (Internet of Things) system designed to prevent children from being forgotten in vehicles. The system is based on a standalone unit installed in the vehicle, combining image processing technologies and biometric sensors to detect human presence after the drive ends. The system provides a double protection envelope: visual motion detection and breathing detection, enabling certain detection even when the child is sleeping or covered. The system interfaces with a dedicated mobile app and sends critical alerts to the parent/driver in case of danger detection.

2. The Need and Problem

Currently, systems for preventing forgotten children exist mainly as an integral part of new and luxury vehicles, or as simple products based on weight alone. Owners of older vehicles are left without a reliable technological answer. Additionally, most existing systems rely on basic alerts inside the vehicle, which might be missed if the driver has already walked away.

Our Solution: A portable, universal system that can be installed in any vehicle via a simple USB connection, providing an active alert directly to the parent's mobile phone.

The uniqueness of the system lies in the use of **"Dual-Verification"** technology: Unlike existing systems relying on a single sensor (only weight or only camera), our system combines two detection elements:

1. **Visual Detection (Camera):** For detecting visible motion.
2. **Biometric Detection (Breathing Sensors):** For detecting a sleeping child who is not moving, or a child located in a "dead zone" (e.g., covered by a blanket or hidden by the safety seat). This combination ensures maximum reliability and minimization of false alarms.

3. System Components

A. The Hardware Unit A compact casing located in the vehicle space containing the following assemblies:

1. **Power Management:**
    - **USB Connection:** The unit connects to the vehicle's USB socket for ongoing charging during the drive.
    - **Internal Backup Battery:** Since most vehicles cut off power supply when the engine is turned off, the unit is provided with a strong internal battery. This battery automatically activates once the vehicle is off, allowing the system to operate, monitor, and transmit alerts even when the vehicle is completely off.
2. **Motion Detection Camera:**
    - Camera capable of detection in varying lighting conditions.
    - **Night Vision:** Use of IR for detection even in absolute darkness.
3. **Breathing Detection Sensors:**
    - Sensitive sensors designed to pick up tiny chest vibrations and breathing sounds. This component is critical for detecting sleeping babies who do not generate large movements picked up by the camera.
4. **Local Alarm:** A speaker/buzzer installed on the unit to sound a loud noise to attract the attention of passersby in the vehicle's surroundings in an emergency.
5. **Controller and Communication Module:** A processor for managing information and a communication component (WiFi/Bluetooth/Sim) for transmitting the alert to the app.

B. Mobile Application The user interface for the parent/driver:

- **Onboarding:** Simple initial connection between the mobile device and the C.A.R.E. unit.
- **Real-Time Critical Alerts:** Upon detection, the app will "wake up" the phone, sounding a loud alarm and vibrating strongly, even if the phone is in "silent" mode.

4. Workflow:

Description of information flow in the system:

1. Activation: The system detects the vehicle stop/engine shutoff and enters "Monitoring Mode" after a predefined delay time (e.g., 60 seconds after the driver exits).
2. Detection Logic:
    - The camera scans the space for motion detection.

    - Simultaneously, sensors sample breathing sounds/vibrations.
    - Data processing is performed to decide if there is human presence.
3. Event Trigger: If a positive detection is found in one or both sensors.
4. Response:
    - Phase A: Sending an urgent alert to the app on the driver's phone.
    - Phase B: (Optional) Activating a loud sound in the unit itself to attract the environment's attention.

Scenarios:

Scenario 1: Routine Exit (Empty Vehicle) Goal: To ensure the system doesn't Send false alerts when everything is fine (preventing false alarms and saving battery).

1. The driver turns off the vehicle (USB power cuts off).
2. The system switches to backup battery and waits a defined time (time to organize for exit).
3. Sensors and cameras scan the space.
4. **Result:** The system does not detect motion and does not detect breathing.
5. **Action:** The system enters "Sleep Mode" to save battery until the next startup

Scenario 2: Forgotten Child Awake (Motion Detection) *Goal: Testing the camera and mobile alert.*

1. The driver turns off the vehicle and exits, but leaves a child in the back seat.
2. After the delay time, the system begins scanning.
3. The camera detects motion in the vehicle.
4. **Action:** The system concludes there is a person in the vehicle despite lack of visual motion.
5. **Immediate Action:** The system sends a command to the app.
6. The unit in the vehicle starts sounding a local beep to attract attention.
7. **User Side:** The parent's phone loud sound and vibrates with a message: "Motion alert in vehicle!".
8. The parent confirms in the app they saw the message and returns to the vehicle.

Scenario 3: Child Forgotten but Not Seen by Camera (Your Special Case - Breathing Detection) Goal: To prove why breathing sensors are needed (child is sleeping or covered).

1. The driver exits the vehicle. The child in the back seat is sleeping and covered with a blanket (the camera does not see them).
2. The system scans: The camera reports "No motion".
3. Simultaneously, breathing sensors identify a steady breathing pattern/vibrations in the space.
4. Action: The system concludes there is a person in the vehicle despite the lack of visual motion.
5. Immediate Action: The system sends a command to the app.
6. The unit in the vehicle starts sounding a local alarm to attract attention.
7. User Side: The parent's phone loud sound and vibrates with a message: "Motion alert in vehicle!".
8. The parent confirms in the app they saw the message and returns to the vehicle.

Scenario 4: Night Scenario (Dark Parking Lot) *Goal: To demonstrate technological capability in difficult conditions.*

1. The vehicle is parked in a completely dark underground parking lot.
2. The system detects a low light level and activates the IR (Infrared) sensors in the camera.
3. **Action:** The system concludes there is a person in the vehicle despite the lack of visual motion (Note: Text repeats standard phrasing).
4. **Immediate Action:** The system sends a command to the app.
5. The unit in the vehicle starts sounding a local alarm to attract attention.
6. **User Side:** The parent's phone loud sound and vibrates with a message: "Motion alert in vehicle!".
7. The parent confirms in the app they saw the message and returns to the vehicle.

System Design - C.A.R.E.

1. High-Level Architecture

The system is built on an IoT Client-Server architecture, consisting of three main layers:

1. Edge Device: The micro-computer in the vehicle performing image processing and sensor reading.
2. Cloud Backend: Serves as a broker for message transmission and user management.
3. Mobile Client: The user application receiving alerts.

2. Component Breakdown

A. The Unit - IoT Side The brain of the system located in the vehicle.

- **Hardware:** Use of Raspberry Pi allowing image processing (specific language not yet decided).
- **Input:** Camera + Breathing Sensor.
- **Local Logic:** The unit does not transmit continuous video to the cloud (to save bandwidth and battery), but processes information locally. Only if a positive identification ("True") occurs, a command is sent to the server.

B. Backend / Cloud

- We plan to search for a suitable cloud service.
- The server's role is to manage authentication and synchronize status in real-time between the vehicle and the phone (Realtime Database).

C. Mobile App

- Cross-platform application to be developed in **React Native** to fit both Android and iOS systems.
- The app listens for changes. Once the status changes to "Danger", the app activates a background service to trigger a critical alert.

3. Database Schema

We will use a Database.

Proposed structure:

Collection: Users

- `user_id` **(Key)**
- `name`**: String**
- `phone_number`**: String**
- `connected_device_id`**: String (FK to Devices)**

Collection: Devices

- `device_id` **(Key - The barcode of the device in the vehicle)**
- `status`**: Enum ["Safe", "Armed", "Danger"]**
- `last_heartbeat`**: Timestamp (To know the system is connected)**
- `battery_level`**: Integer**
- `alerts_history`**: Array**

4. Tools & Libraries

For Image Processing and AI: We have not yet decided which language to use and which libraries we will utilize.

For Communication and App:

- **Firebase SDK:** For managing communication between the Raspberry Pi and the cloud, and between the cloud and the app.
- **FCM (Firebase Cloud Messaging):** Service for managing Push Notifications sent to the phone.

5. Data Flow - Danger Scenario

1. **Detection:** The Raspberry Pi detects motion in the camera or Activity in the breathing sensor.
2. **Processing:** The local algorithm verifies this is not "noise" but a real detection.
3. **Update:** The Raspberry Pi updates the record in Firebase: `status = "Danger"`.
4. **Trigger:** Firebase detects the change and sends a Push Notification to the phone device associated with this ID.
5. **Alert:** The phone beeps.

6. Open Questions

1. **Energy Management:** Since the system needs to operate when the vehicle is off, we are debating the required size of the backup battery and how to optimize the algorithm so the Raspberry Pi doesn't drain the battery too quickly (e.g., using Sleep Mode).
2. **Offline Communication:** How will the system alert if the vehicle is parked in an underground parking lot with no cellular reception (no cloud access)? Should we add a Bluetooth module for direct short-range communication, or rely only on a loud local beep?